

ANATOMY OF AN EMBODIED PHYSICAL AI AGENT

The Tripartite System Hierarchy Across the Proposal–Permission Boundary

1. THE BRAIN · Cognitive Deliberation Tier (MPU / NPU)

High-capacity stochastic neural foundation models, spatial perception, and action chunking running under Linux/POSIX.

Transduction & Sensing

Photons $\rightarrow$ MIPI CSI-2 Ring Buffers

IMU · Encoders · Telemetry

Spatial World Model

ViT Tokens $\rightarrow$ SE(3) Transform Trees

3D Affordance · Latent Belief b_t

Trajectory Deliberation

Diffusion Policy · ACT Action Chunks

Proposes Candidate Trajectory $\hat{u}_{t:t+H}$

PROPOSAL–PERMISSION PRIVILEGE BOUNDARY (Shared SRAM Mailbox · Untrusted $\hat{u}_t \rightarrow$ Verified u^*)

Candidate Proposal $\hat{u}$

2. THE NERVOUS SYSTEM · Real-Time Safety & Timing Tier (MCU)

Isolated bare-metal / RTOS microcontroller executing deterministic 1 kHz safety filters with exclusive actuator authority.

Information Freshness

Hardware Timer Capture

Age Bound $\Delta t \leq \tau_{\max}$ · Freshness Check

Reflex Enforcer (CBF-QP)

Control Barrier Filter $h(\mathbf{x}) \geq 0$

Stopping Envelope $d_{\text{stop}} \leq d_{\text{clear}}$

Actuator Gate Latch

Vetoes Unsafe Proposals $\hat{u}_t$

Latches Certified u^* to PWM Registers

Certified Actuation u^*

3. THE BODY · Continuous Dynamical Plant (Classical Physics)

Physical linkages, power inverters, and motor windings governed by mechanics, electromagnetism, and thermodynamics.

Power Electronics

Gate Drivers · H-Bridge MOSFETs

Phase Currents $I_{\text{phase}} \rightarrow$ Lorentz Torque τ

Kinematics & Dynamics

Momentum $p = mv$ · Contact Stress $F = k\Delta x$

Thermodynamic Joule Heat $\Delta Q = I^2 R \Delta t$

Environment State Mutation

Material World Mutates $W_t \rightarrow W_{t+1}$

Irreversible Work · No Rollback

Endogenous Sensory Shift ($\mathcal{O}_{t+1}$)